

EpiKG: End-to-End Epistemic Preservation in Clinical Knowledge Graphs for Assertion-Aware Retrieval-Augmented Generation

Anonymous Author(s)

Abstract

Clinical NLP pipelines detect assertion status—negation, uncertainty, family history—with $F_1 > 0.90$ [1], yet no existing system propagates these labels through knowledge graph construction into retrieval-augmented generation. We call this the *epistemic propagation gap*: when assertion labels are discarded during graph construction, downstream generation cannot faithfully represent the epistemic state of clinical findings. We present EpiKG, a clinical knowledge graph that closes this gap with a 7-value assertion schema on every edge, a tri-temporal model (valid time, transaction time, and NLP-asserted temporality), and an intent-aware retrieval layer that dispatches question-type-specific graph traversals. On ClinicalBench ($n = 400$, MIMIC-IV, Claude Opus 4.6), intent-aware KG-RAG achieves 76.0% overall (+26 pp vs. LLM alone at 49.8%), improving 8 of 9 categories with zero regressions: change detection rises from 13% to 87% (+73 pp), historical reasoning from 6% to 62% (+56 pp), and family history from 3% to 57% (+53 pp). Brute-force long context fares worse than LLM alone (−5 pp), collapsing to 0% on historical and sequence categories. Cross-model validation reveals that KG-RAG benefit is inversely related to parametric medical knowledge: MedGemma 27B gains only +6 pp from the same pipeline, consistent with a knowledge-equalizer effect. A two-physician blinded adjudication is preregistered. These results establish the category×condition interaction—not the aggregate—as the right evaluation lens for clinical KG-RAG.

1 Introduction

A physician documents “patient denies chest pain, sister had MI at 45, will consider statin if lipids remain elevated”—a single sentence encoding four epistemic states: a negated symptom, a family-history event, a hypothetical action, and an implicit present condition. Clinical NLP can detect these assertions accurately [2, 1], yet when a RAG system ingests the same note, assertion context is flattened: negation is lost, family attribution collapses onto the patient, and downstream reasoning may conclude the patient *has* chest pain and *had* an MI. We call this the *epistemic propagation gap*—the systematic loss of assertion status as clinical information moves from extraction through storage to retrieval.

The gap persists because no system, to our knowledge, propagates assertion labels beyond the extraction stage into knowledge representation or retrieval. OMOP excludes negated conditions from `CONDITION_OCCURRENCE` [3]; FHIR provides `verificationStatus` for conditions only. Graph-augmented RAG systems—GraphRAG [4], GFM-RAG [5], KARE [6], Medical-Graph-RAG [7]—build KGs that discard the metadata distinguishing “patient has diabetes” from “rule out diabetes.” Two KG surveys [8, 9] do not identify assertion preservation as a concern. A parallel *temporal integration gap* exists: clinical events span three time dimensions (valid time, transaction time, NLP-asserted temporality), yet existing systems model at most two [10, 11]. Allen’s interval algebra [12] has been formalized as annotation ontologies [13] and modal operators [14], but never as first-class KG edge attributes that participate in retrieval.

We present EpiKG, a clinical KG system that closes both gaps by preserving epistemic status end-to-end from NLP extraction through OMOP mapping, fact construction, KG materialization, and graph-augmented retrieval. Our contributions:

1. **Epistemic preservation invariant.** We formalize the requirement that assertion status be recoverable at every downstream stage, give an information-theoretic bound on the loss from assertion-blind pipelines (Section 3.5, Appendix B), and realize the invariant with a seven-value schema extending i2b2 [2], carried from extraction through OMOP mapping, fact construction, KG materialization, and RAG serialization.
2. **Tri-temporal KG with Allen’s interval algebra.** Edges carry valid time, transaction time, and NLP-asserted temporality alongside nine temporal relations (seven from Allen’s algebra [12] plus CONCURRENT and UNKNOWN) as first-class attributes, enabling temporal reasoning over longitudinal records.
3. **Five-condition ablation with bookend baselines.** We evaluate five retrieval conditions—from C7 (deterministic KG, no LLM; 3.8%) through C4g (intent-aware KG-RAG; 76.0%, +26 pp vs. LLM alone)—on ClinicalBench and SliceBench. Bookend baselines C6 (brute-force long context; 45.0%) and C7 bracket the design space. Intent-aware KG-RAG improves 8 of 9 categories with zero regressions; neither more context nor structured data alone can substitute for LLM + epistemic KG-RAG.
4. **ClinicalBench, SliceBench, and cross-model validation.** Two benchmarks (400 and 144 questions) across 9 epistemic categories target the propagation gap, with hard longitudinal questions (change, cross-admission, temporal) as the primary endpoint. Cross-model validation on MedGemma 27B and Qwen 35B reveals that KG-RAG benefit is inversely related to parametric medical knowledge; a two-physician blinded adjudication is preregistered (Appendix O).

The central research question is: *under what conditions does structured epistemic context help, hurt, or break even compared to unstructured retrieval or no retrieval at all?* The answer is decisive: intent-aware KG-RAG lifts aggregate accuracy by +26 pp, improving 8 of 9 categories—change detection (+73 pp), historical reasoning (+56 pp), family-history attribution (+53 pp)—with zero regressions. Brute-force long context (−5 pp) performs *worse* than no retrieval at all. The benefit is model-dependent: MedGemma 27B gains only +6 pp from the same pipeline, with a historical regression (−46 pp) where it already performs well—suggesting that KG-RAG benefit is inversely related to parametric medical knowledge. The magnitude and direction of the effect are both category- and model-specific, producing interactions that aggregate accuracy alone obscures (Section 5).

2 Related Work

We organize prior work along four axes and synthesize the gaps that EpiKG addresses (Table 7, Appendix).

Medical RAG systems. Graph-augmented retrieval has emerged as a leading paradigm for medical QA. GraphRAG [4] introduced community-based summarization; GFM-RAG [5] trained an 8M-parameter graph foundation model across 60 KGs (14M+ triples); KARE [6] adapted community retrieval to clinical decision support; DoctorRAG [15] emulates physician reasoning; MedRAG [16] constructs hierarchical diagnostic KGs; Medical-Graph-RAG [7] links documents via triple graphs. Yet no published medical RAG system models assertion status or temporal context; GFM-RAG and KARE build population-level graphs rather than patient-level graphs from clinical text, and a recent survey [17] confirms no system incorporates epistemic metadata into the retrieval graph.

Clinical KG construction. Multi-LLM KG-RAG [18] uses multi-agent LLM prompting with schema-constrained extraction for oncology, modeling uncertainty via entropy but not tracking assertion classes on edges. AutoRD [19] applies GPT-4 to rare disease literature; RECAP-KG [20] handles messy GP notes; FHIR-Ontop-OMOP [21] generates virtual KGs from OMOP data. All share a common limitation: assertion status is not propagated into the final graph.

Assertion detection. NegEx [22] introduced trigger-based negation detection; ConText [23] extended it with temporality and experienter. The i2b2/VA shared task [2] formalized a six-class taxonomy; Gul et al. [1] fine-tuned LLMs to 0.962 accuracy. All treat assertion detection as a terminal annotation task: labels are not carried into knowledge graphs or retrieval systems.

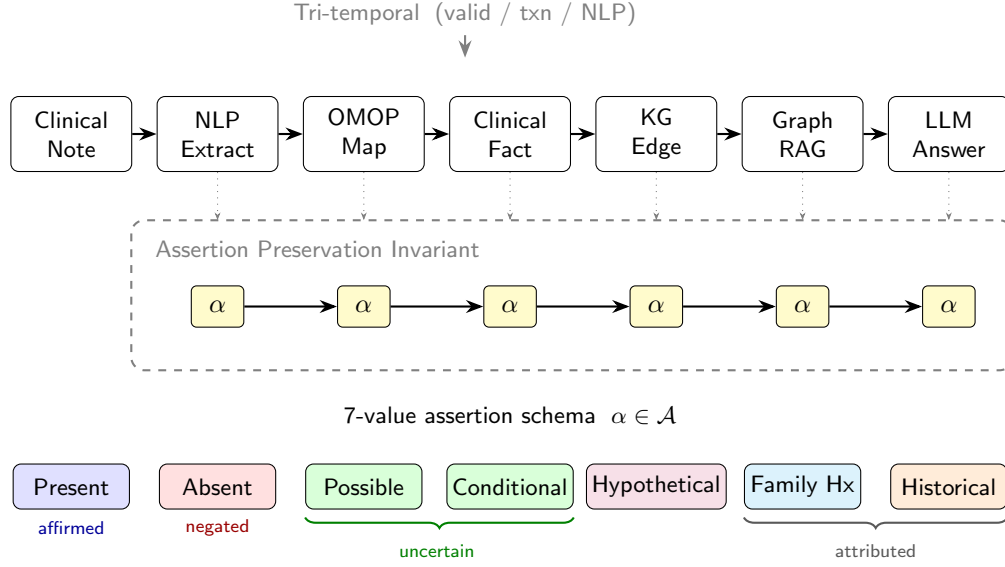


Figure 1: The EpiKG pipeline. The assertion label α (highlighted) is preserved as a first-class attribute from NLP extraction through OMOP mapping, fact construction, KG materialization, and assertion-aware GraphRAG.

Assertion detection is necessary but not sufficient: labels must also be *temporally situated*—the same condition may be present in one encounter and absent in the next.

Temporal knowledge graphs. MedTKG [10] constructs temporal KGs with time-stamped snapshots (event time only); Graphiti [11] implements bitemporal edges [24] but lacks clinical ontology alignment; TEO [13] and TCL [14] formalize Allen’s relations [12] as annotation ontologies or modal operators rather than KG edge attributes. Uncertainty quantification in biomedical KGs [25] has not been connected to assertion taxonomies. Two structural gaps emerge: an *epistemic propagation gap* (assertion-detecting systems do not persist labels into KGs) and a *temporal integration gap* (temporal formalisms operate as annotation layers, not as KG edge attributes participating in retrieval). EpiKG closes both by carrying assertion and temporal metadata as first-class properties through every pipeline stage.

3 System Design

3.1 Overview

EpiKG transforms unstructured clinical narratives into an assertion-aware, temporally grounded knowledge graph exposed through graph-augmented retrieval. The pipeline comprises six stages: (1) document ingestion, (2) NLP extraction, (3) OMOP concept normalization [3], (4) clinical fact construction with epistemic metadata, (5) KG materialization with shared concept nodes, and (6) assertion-aware retrieval. The key invariant is that *epistemic assertion status* is a first-class attribute at every stage, not an NLP annotation discarded after extraction (Figure 1). Clinical concepts are stored once as shared global nodes; patient-specific semantics (assertion status, temporality, confidence) reside on edges, enabling cohort queries via edge scans.

3.2 Epistemic Assertion Schema

Clinical notes abound with qualified statements—“*no evidence of pneumonia*,” “*possible CHF*,” “*mother had breast cancer*”—yet standard representations discard these qualifiers. OMOP excludes negated conditions entirely [3]; FHIR limits assertion metadata to conditions. EpiKG defines a seven-value assertion taxonomy:

$$\alpha \in \{\text{Pres.}, \text{Abs.}, \text{Poss.}, \text{Cond.}, \text{Hypo.}, \text{Fam.Hx.}, \text{Hist.}\} \quad (1)$$

This extends the i2b2 six-class taxonomy [2] by separating HISTORICAL from FAMILY_HISTORY, a clinically important distinction that prior work conflates (definitions in Appendix Q). A rule-based classifier (101 scope-aware trigger patterns) assigns α with an associated confidence score; we chose rule-based classification for deterministic behavior and interpretability, though fine-tuned LLM classifiers [1] could be substituted. Intrinsic classifier evaluation is deferred; the end-to-end Clinical-Bench ablation (C3→C4) serves as a functional test of assertion propagation. The label is carried through every stage: from ExtractedMention through ensemble merging, ClinicalFact records, KG edge properties, and retrieval scoring. To our knowledge, no prior system [1, 26] maintains this end-to-end invariant.

3.3 Tri-Temporal Knowledge Graph Model

Clinical events unfold across multiple time dimensions that prior systems model incompletely [10, 11]. Bi-temporal databases [24] distinguish valid time from transaction time; EpiKG adds a third dimension (NLP-asserted temporality) and stores all three on every KG edge:

Definition 1 (Tri-Temporal Edge). *An edge $e = (s, p, o, \alpha, \tau_v, \tau_t, \tau_a, r, c)$ where:*

- s, o are source and target nodes; p is the predicate (one of 24 edge types);
- $\alpha \in \mathcal{A}$ is the assertion label (Eq. 1);
- $\tau_v = (\text{event_date}, \text{valid_from}, \text{valid_to})$ is the **valid time** interval—when the relationship held in the real world;
- $\tau_t = (\text{recorded_at}, \text{doc_date}, \text{created_at})$ is the **transaction time**—when it was recorded;
- $\tau_a \in \{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$ is the **NLP-asserted temporality**;
- $r \in \mathcal{R}$ is an Allen interval algebra relation [12]; $c \in [0, 1]$ is temporal confidence.

$\mathcal{R}$ comprises seven of Allen’s 13 canonical relations [12] (merging symmetric pairs like Before/Meets) plus CONCURRENT and UNKNOWN (nine total; full mapping in Appendix R). Unlike TEO [13] and TCL [14], which use Allen’s relations as annotation-ontology classes or modal operators, EpiKG stores r and c directly on materialized edges, enabling efficient temporal filtering during traversal.

3.4 Assertion-Aware Graph-Augmented Retrieval

Given a clinical question q and patient π , the retrieval pipeline: (1) extracts concepts from q via NLP + OMOP enrichment; (2) traverses 2–3 hops via bidirectional BFS over patient KG edges and OMOP vocabulary relationships (20M+ edges) via PostgreSQL CTEs (Appendix K), pruning edges with $c < 0.3$; (3) groups edges into four temporal views (event timeline, current state, historical, conflicts); (4) retrieves matching guideline sections; (5) scores edges by:

$$\text{score}(e) = c(e) + \mathbb{I}[\text{type}(e) \in Q_{\text{rel}}] \cdot 0.2 + \mathbb{I}[\tau_a(e) = \text{CURRENT}] \cdot 0.1 \quad (2)$$

where the bonuses for question-relevant edge types (0.2) and current temporality (0.1) were set by manual tuning on a development set of 20 questions. The surviving subgraph is serialized into structured text preserving assertion labels (e.g., “ABSENT: pneumonia”); and (6) composes graph evidence, temporal context, guidelines, and source documents into a single prompt.

3.5 Formal Epistemic Preservation

We formalize the epistemic invariant maintained by the EpiKG pipeline, building on the ontological principle that aboutness must be preserved across transformations [27]. The *epistemic state* of a clinical mention m is the tuple $e(m) = (c, \alpha, \xi, \tau)$: OMOP concept, 7-value assertion label, experiencer ($\{\text{PATIENT}, \text{FAMILY}\}$), and temporality ($\{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$); see Table 6 (Appendix) for notation summary. A pipeline *epistemically preserves* m if the output epistemic state equals the extraction state at every stage. We formalize (Appendix B) that an assertion-blind pipeline collapses the assertion distribution to a degenerate point mass, incurring entropy loss $\Delta H = H(A_c^\pi) \geq 0$ [28], and that assertion-faithful accuracy is bounded by $1 - f_{np}(c)$, where f_{np} is the fraction of non-present assertions. The empirical consequences are measured via category-stratified accuracy in Section 5.

Table 1: ClinicalBench conditions. C1/C4/C4g form the ablation ladder; C7 and C6 are bookend baselines.

ID	Condition	Retrieval	Assertion	Temporal
C7	Deterministic KG	KG lookup (no LLM)	Full	Tri-temporal
C1	LLM Alone	None	None	None
C4	+ Epistemic KG-RAG	Graph + Doc	Full (7-class)	Tri-temporal
C4g	+ Intent-Aware KG-RAG	Graph + Doc (type-specific)	Full (7-class)	Tri-temporal
C6	Long Context	All notes	None	None

3.6 Intent-Aware Retrieval (C4g)

The multi-hop BFS described above treats all questions uniformly, yet different clinical question types require fundamentally different graph operations. A *change* question requires cross-admission set differencing; a *current-state* question needs filtering to the most recent valid edges; a *historical* question must recover resolved conditions.

Intent classifier. A lightweight rule-based classifier maps each question to one of four intents—CHANGE, CURRENT_STATE, HISTORICAL, or DEFAULT—using question metadata with keyword-pattern fallback (Algorithm 1, Appendix S).

Change retrieval. For $\iota = \text{CHANGE}$, edges are partitioned by admission (`hadm_id`). The module computes set differences across admission pairs (k, k') —concepts *added* $(\mathcal{C}_{k'} \setminus \mathcal{C}_k)$, *removed* $(\mathcal{C}_k \setminus \mathcal{C}_{k'})$, and *continued* $(\mathcal{C}_k \cap \mathcal{C}_{k'})$ —and returns structured evidence.

Current-state and historical retrieval. For CURRENT_STATE, the module filters to edges with $\tau_a = \text{CURRENT}$ or open-ended validity, deduplicates by concept, and emits “NOT FOUND” for missing concepts. For HISTORICAL, it selects edges with $\tau_a = \text{PAST}$ and augments with admission-based inference (concepts in earlier admissions but absent from the latest are labeled “resolved”). Each intent triggers a type-specific prompt template that structures evidence according to question semantics (Appendix U).

4 Benchmark Design

Existing clinical QA benchmarks evaluate factual recall (MedQA [29]) or agent task completion [30] but do not test handling of epistemic qualifiers or reasoning over longitudinal records of varying complexity. We introduce two complementary benchmarks built on MIMIC-IV [31].

4.1 ClinicalBench: Assertion-Sensitive Clinical QA

ClinicalBench comprises 400 gold-standard questions over 45 MIMIC-IV patients in two tasks: **Task A** (200 questions, negation-aware retrieval) and **Task B** (200 questions, temporal reasoning). Questions span 9 categories: *negation*, *conditional*, *uncertainty*, *family_history*, *sequence*, *current_state*, *duration*, *historical*, and *change*. Gold-standard answers were created through manual chart review and verified against source documents; a pilot physician adjudication ($n = 5$, Appendix N) assessed quality. Three ablation conditions progressively add system components (Table 1). Two bookend baselines—C6 (all notes concatenated) and C7 (deterministic KG lookup, no LLM)—bracket the design space.

Endpoints. The **pre-registered primary endpoint** is the hard longitudinal subset (*change* + *current_state* + *historical*, $n = 130$): C4g vs. C1 aggregate accuracy. The **secondary endpoint** is full 400-question ClinicalBench, C4g vs. C1. **Diagnostic endpoints** are per-category C1→C4g accuracy deltas and the C4→C4g transition on the hard subset.

What each transition isolates. C1→C4 isolates the *epistemic retrieval effect*: C4 adds graph traversal with assertion metadata, testing whether structured context helps or hurts. C4→C4g isolates the *intent-aware routing effect*: both carry full assertion metadata, but C4g dispatches question-type-specific graph traversals. C4g vs. C6 isolates the *structured-vs-unstructured retrieval effect* on the same questions.

4.2 SliceBench: Complexity-Stratified Longitudinal QA

SliceBench tests the hypothesis that KG-augmented retrieval provides increasing benefit as patient record complexity grows. We select 6 MIMIC-IV patients in three tiers: **Tier A** (2 patients,

Table 2: ClinicalBench primary results (400 questions, Claude Opus 4.6, keyword evaluator). C7 is model-independent (no LLM); remaining conditions use Opus 4.6. Best in **bold**.

	Condition	Accuracy	Δ vs C1
C7	Deterministic KG (no LLM)	3.8%	−46.0 pp
C1	LLM Alone	49.8%	—
C4	+ Epistemic KG-RAG	71.5%	+21.7 pp
C4g	+ Intent-Aware KG-RAG	76.0%	+26.3 pp
C6	Long Context (all notes)	45.0%	−4.8 pp

1–2 encounters), **Tier B** (2 patients, 5–10), and **Tier C** (2 patients, 15+). Each patient receives 24 questions (including hard longitudinal categories: cross-encounter medication timelines, problem list reconciliation, causal chain tracing), yielding 144 total. Five conditions (B0–B4) form a monotone context progression from parametric-only to full system (Appendix J). The critical comparison is B2→B3: both have the same documents, but B3 adds structured KG context with assertion metadata and temporal relations.

4.3 Evaluation Protocol

ClinicalBench uses a *deterministic keyword evaluator* that performs exact word-boundary matching of gold-standard assertion and temporal keywords against model answers (keyword lists in Appendix P). This evaluator is fully reproducible (no stochastic LLM judge) and is used for all ClinicalBench results. The primary answering model is Claude Opus 4.6; cross-model validation uses MedGemma 27B and Qwen 35B under the same evaluator.

SliceBench uses an *LLM-as-judge* protocol with separated answering and judging models to prevent self-evaluation bias [32]: Claude Sonnet 4.5 generates answers and Claude Opus 4.6 judges correctness.

Statistical reporting. We report BCa bootstrap 95% CIs ($n = 2,000$, seed 42) on SliceBench aggregate metrics (Table 5); CIs are computed at the question level. ClinicalBench results are reported as point estimates with within-run paired comparisons; CIs are omitted because the primary analysis targets condition deltas rather than absolute levels. A safety score with asymmetric weighting ($w = 2.0$ for false-positive assertion errors) captures clinical risk (Appendix D).

4.4 Physician Validation Protocol

To validate automated scoring, we preregister a two-physician blinded adjudication. Two board-certified emergency medicine physicians independently score a stratified sample of 168 questions under two conditions (C1 vs. C4g), blinded to condition assignment, using the same binary correctness rubric as the keyword evaluator. A third physician adjudicates disagreements; inter-rater reliability is reported as Cohen’s κ . The protocol (Appendix O) was finalized before examining primary results. Results are pending.

5 Results

We evaluate EpiKG through two benchmarks: ClinicalBench tests assertion-sensitive and longitudinal reasoning, while SliceBench measures longitudinal QA across patient complexity tiers. ClinicalBench uses a deterministic keyword evaluator for reproducibility; SliceBench uses LLM-as-judge evaluation (Section 4.3). Claude Opus 4.6 is the primary answering model; MedGemma 27B and Qwen 35B provide cross-model validation. SliceBench reports BCa bootstrap 95% CIs ($n = 2,000$, seed 42) [33], resampled at the question level.

5.1 ClinicalBench: Primary Ablation

The bookend baselines bracket the design space: C7 (deterministic KG, no LLM) achieves 3.8%—only negation detection works (13.6%) via stored assertion metadata—confirming that an LLM is essential for reasoning over graph evidence. C6 (all notes concatenated) scores 45.0%, *worse* than C1 (−4.8 pp): brute-force long context is not a viable alternative, even with Opus’s 200K token window. Uniform epistemic KG-RAG (C4) achieves 71.5% (+21.7 pp), showing that structured retrieval with

Table 3: ClinicalBench per-category accuracy (%) across conditions (Claude Opus 4.6). C4g improves 8 of 9 categories over C1 with zero regressions. Best per row in **bold**.

Category	n	C6	C1	C4	C4g	C4g−C1
Negation	110	72.7	86.4	100.0	88.2	↑ 1.8 pp
Conditional	20	45.0	15.0	70.0	45.0	↑ 30.0 pp
Uncertainty	40	30.0	17.5	57.5	50.0	↑ 32.5 pp
Family hist.	30	43.3	3.3	70.0	56.7	↑ 53.3 pp
Sequence	40	0.0	100.0	87.5	100.0	= 0 pp
Current st.	50	48.0	40.0	54.0	72.0	↑ 32.0 pp
Duration	30	73.3	86.7	83.3	93.3	↑ 6.7 pp
Historical	50	0.0	6.0	60.0	62.0	↑ 56.0 pp
Change	30	66.7	13.3	3.3	86.7	↑ 73.3 pp
Overall	400	45.0	49.8	71.5	76.0	↑ 26.3 pp



Figure 2: ClinicalBench per-category accuracy across four conditions (C6, C1, C4, C4g; Claude Opus 4.6). C6 (orange dash-dot) collapses on historical and sequence (both 0%); C4 (green dashed) improves most categories but drops change to 3%; C4g (dark solid) improves 8 of 9 categories over C1 with no regressions.

assertion metadata provides substantial benefit even without intent-aware routing. Intent-aware KG-RAG (C4g) adds another +4.5 pp to reach 76.0%. The C4→C4g transition is most visible on change detection: C4 drops to 3.3% (worse than C1’s 13.3%) because uniform retrieval injects irrelevant temporal context, while C4g recovers to 86.7% via question-type-specific traversal.

5.2 Category×Condition Interaction

Table 3 and Figure 2 reveal a strong *category×condition interaction*: intent-aware KG-RAG improves 8 of 9 categories with zero regressions—sequence is already at 100% under C1.

Table 4: ClinicalBench cross-model results (400 questions, keyword evaluator). KG-RAG benefit is inversely related to parametric medical knowledge.

	Model	C1	C4g	Δ
Commercial	Claude Opus 4.6	49.8%	76.0%	+26.3 pp
Open	MedGemma 27B	59.9%	66.0%	+6.1 pp
Open	Qwen 35B	—%	—%	— pp

Table 5: SliceBench overall results (144 questions, Claude Sonnet 4.5, judged by Opus 4.6). Sig: * denotes CI excluding zero.

	Condition	Score	95% CI	Δ vs B0
B0	LLM Alone	49.9%	[43.6%, 56.1%]	—
B1	Latest Note	73.4%	[67.8%, 78.7%]	+23.5 pp
B2	All Notes RAG	84.7%	[80.2%, 88.8%]	+34.8 pp *
B3	KG-RAG	86.9%	[82.8%, 90.8%]	+37.0 pp *
B4	Full System	87.6%	[83.7%, 91.2%]	+37.8 pp *

246 **Where intent-aware routing excels.** The largest gains occur on categories requiring cross-
 247 admission synthesis: change (+73 pp), historical (+56 pp), family history (+53 pp), uncertainty
 248 (+33 pp), and current state (+32 pp). C6 fails catastrophically on historical (0%) and sequence (0%)
 249 despite Opus’s 200K window, confirming that long context cannot substitute for structured retrieval.

250 5.3 Cross-Model Validation

251 The KG-RAG benefit is model-dependent and inversely related to parametric medical knowledge
 252 (Table 4). MedGemma 27B, fine-tuned on medical corpora, achieves a higher C1 baseline (59.9% vs.
 253 49.8%) but gains only +6.1 pp from KG-RAG. Opus, a general-purpose model with lower medical
 254 baseline, gains +26.3 pp—KG-RAG compensates for the knowledge gap.

255 This pattern is most visible on historical questions: MedGemma achieves 86% on C1 (it identifies
 256 historical status parametrically) but *regresses* under MedGemma C4g (−46 pp) because structured
 257 retrieval formatting overrides correct parametric reasoning. Opus starts at 6% on C1 and gains
 258 +56 pp under C4g—the epistemic KG provides knowledge the model lacks. Intent-aware KG-RAG
 259 thus functions as a *medical knowledge equalizer*, providing the largest benefit to models that need it
 260 most. Per-category cross-model breakdowns appear in Appendix Table 8.

261 5.4 SliceBench

262 The incremental KG layer (B2→B3) contributes +2.2 pp overall (CI: [−1.5 pp, +5.9 pp]), not
 263 reaching aggregate significance, but tier-stratified estimates show a complexity-dependent trend:
 264 +0.6 pp (Tier A), +1.0 pp (Tier B), +5.0 pp (Tier C, 15+ encounters), consistent with KG context
 265 being most valuable when document volume overwhelms chunk-level retrieval (Appendix L).

266 5.5 Physician Adjudication

267 A two-physician blinded adjudication ($n = 168$ questions $\times$ 2 conditions, Cohen’s κ for inter-
 268 rater reliability) is preregistered to validate automated scoring; the protocol (Appendix O) was de-
 269 signed before examining results. *Results are pending.*

270 6 Discussion and Limitations

271 **Why long context fails.** Brute-force long context (C6) scores −4.8 pp below LLM alone, even
 272 with Opus’s 200K token window. C6 catastrophically fails on historical (0%) and sequence (0%):
 273 concatenating all notes chronologically buries temporal distinctions that are critical for these cat-
 274 egories. Historical questions require distinguishing resolved from active conditions across admis-
 275 sions; sequence questions require strict temporal ordering—both are implicit in raw text but ex-
 276 plicit in the epistemic KG. C6 handles change detection reasonably (67%) because the relevant
 277 information appears explicitly in the text, but KG-RAG still outperforms it (87%, +20 pp). Epis-
 278 temic KG-RAG without intent-awareness (C4, MedGemma run) also hurts (−4.9 pp), consistent
 279 with GraphRAG-Bench [34]: uniform graph traversal injects context that can override correct para-
 280 metric reasoning.

Model-dependent KG benefit. The most striking finding is that KG-RAG benefit is inversely related to parametric medical knowledge (Table 4). Opus, a general-purpose model, gains +26 pp from intent-aware KG-RAG; MedGemma 27B, fine-tuned on medical corpora, gains only +6 pp. The mechanism is clearest on historical questions: MedGemma achieves 86% on C1—it identifies historical conditions from textual cues parametrically—but *regresses* −46 pp under C4g because structured retrieval formatting overrides its correct reasoning. Opus starts at 6% on C1 (it lacks this parametric capability) and gains +56 pp from the same retrieval pipeline. Intent-aware KG-RAG thus functions as a *medical knowledge equalizer*: the structured epistemic context compensates for knowledge the model lacks, while models that already possess the knowledge gain less or are actively harmed by redundant context. This has practical implications: the same RAG system may help or hurt depending on the downstream model, and evaluation on a single model understates or overstates the system’s value.

When KG context matters. The complexity-dependent SliceBench effect (Tier C: +5.0 pp vs. Tier A: +0.6 pp) suggests structured epistemic context is most valuable when: (a) document volume overwhelms chunk-level retrieval, (b) questions require cross-encounter synthesis where the same concept appears with different assertion states, and (c) assertion status evolves across encounters. C4g’s intent-aware routing provides direct evidence: the “change” category jumps from 13% (C1) to 87% (+73 pp), confirming that *targeted graph traversal strategy*, not text access, is the key factor.

Uncertainty remains weakest. Uncertainty improves from 17.5% to 50.0% (+32.5 pp), but absolute accuracy remains low. Possible and suspected conditions are linguistically subtle (“consider statin,” “may have CHF”) and require pragmatic inference beyond keyword matching.

Limitations. Sample size is modest: ClinicalBench covers 45 patients (400 questions) and SliceBench 6 patients (144 questions), both from MIMIC-IV [31]. The B2→B3 CI crosses zero ([-1.5 pp, +5.9 pp]); tier-level estimates ($n = 2$ patients per tier) are hypothesis-generating only. Model-dependent effects mean headline gains (+26 pp) are specific to Opus + C4g; MedGemma shows +6 pp with a −46 pp historical regression. The keyword evaluator may miss semantically correct answers; physician adjudication (Section 5.5) will address this. This is an ablation study; comparisons with medical RAG systems [5, 6, 15] were not performed because these target factoid QA rather than patient-level longitudinal reasoning. Bootstrap CIs are question-level; patient-level clustering is planned. MedGemma run-to-run variance (± 10 pp) arises from GPU non-determinism in 4-bit GGUF inference. Over-reliance on KG-driven summaries risks anchoring bias if NLP extraction introduces systematic errors; single-site evaluation (US academic center) leaves generalization unvalidated (Appendix V).

Conclusion. The epistemic propagation gap is a real and measurable phenomenon: brute-force long context degrades accuracy by 5 pp, while intent-aware epistemic KG-RAG improves it by 26 pp. EpiKG addresses this gap through end-to-end epistemic preservation combined with intent-aware retrieval that matches graph traversal strategy to question semantics. The benefit is category-specific: change detection rises from 13% to 87% (+73 pp), historical from 6% to 62% (+56 pp), family history from 3% to 57% (+53 pp)—8 of 9 categories improve with zero regressions. The benefit is also model-dependent: KG-RAG functions as a medical knowledge equalizer, providing the largest gains to models that need it most. Neither more context nor structured data alone suffices—C6 and C7 (3.8%) bracket the design space—confirming that the category×condition interaction, not the aggregate, is the right lens for clinical KG-RAG evaluation.

References

- [1] Veysel Kocaman, Yigit Gul, M. Aytug Kaya, Hasham Ul Haq, Mehmet Butgul, Cabir Celik, and David Talby. Beyond negation detection: Comprehensive assertion detection models for clinical NLP. In *Text2Story Workshop at European Conference on Information Retrieval (ECIR)*, 2025.
- [2] Özlem Uzuner, Brett R. South, Shuying Shen, and Scott L. DuVall. 2010 i2b2/VA challenge on concepts, assertions, and relations in clinical text. *Journal of the American Medical Informatics Association*, 18(5):552–556, 2011.

- [3] OHDSI Collaborative. OMOP common data model v5.4. *Observational Health Data Sciences and Informatics*, 2024.
- [4] Darren Edge et al. From local to global: A graph RAG approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*, 2024.
- [5] Tianjun Luo et al. GFM-RAG: Graph foundation model for retrieval augmented generation. In *Advances in Neural Information Processing Systems*, 2025.
- [6] Peng Jiang et al. KARE: Knowledge graph augmented reasoning via llms for clinical decision support. In *International Conference on Learning Representations*, 2025.
- [7] Junde Wu et al. Medical-Graph-RAG: Towards safe medical large language model via graph retrieval-augmented generation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, 2025.
- [8] Hassan S. Al Khatib, Subash Neupane, Harish Kumar Manchukonda, Noorbakhsh Amiri Gollarz, Sudip Mittal, Amin Amirlatifi, and Shahram Rahimi. Patient-centric knowledge graphs: A survey of current methods, challenges, and applications. *Frontiers in Artificial Intelligence*, 7:1388479, 2024. doi: 10.3389/frai.2024.1388479.
- [9] Bilal Abu-Salih, Muhammad Al-Qurishi, Mohammed Alweshah, Mohammad Al-Smadi, Reem Alfayez, and Heba Saadeh. Healthcare knowledge graph construction: A systematic review of the state-of-the-art, open issues, and opportunities. *Journal of Big Data*, 10(1):81, 2023. doi: 10.1186/s40537-023-00774-9.
- [10] Marco Postiglione, Daniel Bean, Zeljko Kraljevic, Richard Dobson, and Vincenzo Moscato. Predicting future disorders via temporal knowledge graphs and medical ontologies. *IEEE Journal of Biomedical and Health Informatics*, 28(7):4238–4248, 2024. doi: 10.1109/JBHI.2024.3390419.
- [11] Daniel Ramirez et al. Graphiti: Building real-time knowledge graphs with temporal awareness. *arXiv preprint arXiv:2501.13956*, 2025.
- [12] James F. Allen. Maintaining knowledge about temporal intervals. *Communications of the ACM*, 26(11):832–843, 1983.
- [13] Fei Li, Jianfu Hong, Cui Tao, et al. TEO: A time event ontology for clinical narratives. *Journal of the American Medical Informatics Association*, 27(10):1560–1568, 2020.
- [14] Yan Huang, Xiaojin Li, and Guo-Qiang Zhang. Temporal cohort logic. *AMIA Annual Symposium Proceedings*, 2022:1237–1246, 2023.
- [15] Yifan Lu, Tianyu Fu, et al. Doctorrag: Medical rag emulating doctor-like reasoning. In *Advances in Neural Information Processing Systems*, 2025.
- [16] Yusheng Wang et al. MedRAG: Enhancing medical diagnosis through retrieval-augmented generation with knowledge graph-elicited reasoning. *Proceedings of The Web Conference*, 2025.
- [17] Boci Peng, Yun Zhu, Yongchao Liu, Xiaohe Bo, Haizhou Shi, Chuntao Hong, Yan Zhang, and Siliang Tang. Graph retrieval-augmented generation: A survey. *ACM Transactions on Information Systems*, 2025. doi: 10.1145/3777378.
- [18] Wei Chen et al. Multi-LLM KG-RAG: End-to-end clinical knowledge graph construction. *arXiv preprint arXiv:2601.01844*, 2026.
- [19] Lang Li et al. AutoRD: An automatic and end-to-end system for rare disease knowledge graph construction. *JMIR Medical Informatics*, 12, 2024.
- [20] Rakhilya Lee Mekhtieva, Brandon Forbes, Dalal Alrajeh, Brendan Delaney, and Alessandra Russo. RECAP-KG: Mining knowledge graphs from raw GP notes for remote COVID-19 assessment in primary care. *arXiv preprint arXiv:2306.17175*, 2023.
- [21] Guohui Xiao, Emily Pfaff, Eric Prud’hommeaux, David Booth, Deepak K. Sharma, Nan Huo, Yue Yu, Nansu Zong, Kathryn J. Ruddy, Christopher G. Chute, and Guoqian Jiang. FHIR-Ontop-OMOP: Building clinical knowledge graphs in FHIR RDF with the OMOP common data model. *Journal of Biomedical Informatics*, 134:104201, 2022. doi: 10.1016/j.jbi.2022.104201.

- [22] Wendy W. Chapman, Will Bridewell, Paul Hanbury, Gregory F. Cooper, and Bruce G. Buchanan. A simple algorithm for identifying negated findings and diseases in discharge summaries. *Journal of Biomedical Informatics*, 34(5):301–310, 2001.
- [23] Henk Harkema, John N. Dowling, Tyler Thornblade, and Wendy W. Chapman. ConText: An algorithm for determining negation, experienter, and temporal status from clinical reports. *Journal of Biomedical Informatics*, 42(5):839–851, 2009.
- [24] Richard T. Snodgrass. *Developing Time-Oriented Database Applications in SQL*. Morgan Kaufmann, 2000.
- [25] Adil Bahaj and Mounir Ghogho. A step towards quantifying, modelling and exploring uncertainty in biomedical knowledge graphs. *Computers in Biology and Medicine*, 184:109355, 2025. doi: 10.1016/j.compbimed.2024.109355.
- [26] Yifan Peng, Xiaosong Wang, Le Lu, Mohammadhadi Bagheri, Ronald Summers, and Zhiyong Lu. NegBio: A high-performance tool for negation and uncertainty detection in radiology reports. *AMIA Summits on Translational Science Proceedings*, 2018.
- [27] Werner Ceusters and Barry Smith. Aboutness: Towards foundations for the information artifact ontology. *International Conference on Biomedical Ontology*, 2015.
- [28] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948.
- [29] Di Jin, Eileen Pan, Nassim Oufattole, Wei-Hung Weng, Hanyi Fang, and Peter Szolovits. What disease does this patient have? a large-scale open domain question answering dataset from medical exams. *Applied Sciences*, 11(14):6421, 2021. doi: 10.3390/app11146421.
- [30] Yixing Jiang, Kameron C. Black, Gloria Geng, Danny Park, James Zou, Andrew Y. Ng, and Jonathan H. Chen. MedAgentBench: A virtual EHR environment to benchmark medical LLM agents. *NEJM AI*, 2(9), 2025. doi: 10.1056/AIdbp2500144.
- [31] Alistair E.W. Johnson, Lucas Bulgarelli, Lu Shen, et al. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, 10:1, 2023.
- [32] Guangzhi Xiong et al. MIRAGE: Medical information retrieval-augmented generation evaluation. In *Findings of the Association for Computational Linguistics: ACL*, 2024.
- [33] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall/CRC, 1993.
- [34] Yilin Xiao, Junnan Dong, Chuang Zhou, Su Dong, Qian-wen Zhang, Di Yin, Xing Sun, and Xiao Huang. GraphRAG-Bench: Challenging domain-specific reasoning for evaluating graph retrieval-augmented generation. In *International Conference on Learning Representations*, 2026.
- [35] Yanjun Gao, Ruizhe Li, Emma Croxford, John Caskey, Brian W. Patterson, Matthew Churpek, Timothy Miller, Dmitriy Dligach, and Majid Afshar. Leveraging medical knowledge graphs into large language models for diagnosis prediction: Design and application study. *JMIR AI*, 4(1):e58670, 2025. doi: 10.2196/58670.

421 A Notation Summary

Symbol	Meaning
$\alpha \in \mathcal{A}$	Assertion label (7 values, Eq. 1)
τ_v	Valid time (event date, valid from/to)
τ_t	Transaction time (recorded at, doc date, created at)
τ_a	NLP-asserted temporality (Current / Past / Future)
$r \in \mathcal{R}$	Temporal relation (9 values, Table 16)
c	Confidence score $\in [0, 1]$
ξ	Experiencer (Patient / Family)
$e(m)$	Epistemic state tuple (c, α, ξ, τ)
q, π	Clinical question, patient
ι	Question intent (Change / Current_State / Historical / Default)
$f_{np}(c)$	Fraction of non-present assertions for concept c

Table 6: Key notation used throughout the paper.

422 B Formal Epistemic Preservation

423 We formalize the epistemic invariant maintained by the EpiKG pipeline, building on the principle
 424 that aboutness—the relationship between information artifacts and the entities they represent—must
 425 be preserved across transformations [27].

426 **Definition 2** (Epistemic State). *For a clinical mention m , the epistemic state is the tuple $e(m) =$
 427 (c, α, ξ, τ) , where c is the OMOP concept identifier, $\alpha \in \mathcal{A}$ is the 7-value assertion label (Eq. 1), $\xi \in$
 428 $\{\text{PATIENT, FAMILY}\}$ is the experiencer, and $\tau \in \{\text{CURRENT, PAST, FUTURE}\}$ is the temporality. A
 429 pipeline P epistemically preserves mention m if $P(e(m)) = e(m)$; that is, the epistemic state at the
 430 output of every pipeline stage is identical to the state at extraction.*

Proposition 3 (Assertion Entropy Loss). *For concept c in patient π ’s record, let A_c^π denote the
 assertion distribution with empirical frequencies $p(\alpha_i) = n_i / \sum_j n_j$ over the $|\mathcal{A}|$ assertion classes.
 The assertion entropy is*

$$H(A_c^\pi) = - \sum_i p(\alpha_i) \log p(\alpha_i) . \quad (3)$$

431 *An assertion-blind pipeline collapses all mentions to $\alpha = \text{PRESENT}$, yielding a degenerate distribu-*
 432 *tion with $H = 0$. The information loss is $\Delta H = H(A_c^\pi) \geq 0$ [28], with $\Delta H > 0$ strictly whenever*
 433 *any mention carries a non-present assertion.*

434 *Proof.* Under the assertion-blind mapping $\phi : \alpha_i \mapsto \text{PRESENT}$ for all i , the output distribution
 435 assigns probability 1 to PRESENT and 0 to all other classes, so $H(\phi(A_c^\pi)) = 0$. By non-negativity
 436 of Shannon entropy, $\Delta H = H(A_c^\pi) - 0 = H(A_c^\pi) \geq 0$, with equality iff all mentions already carry
 437 $\alpha = \text{PRESENT}$. $\square$

438 **Corollary 4** (Faithfulness Bound). *Let $f_{np}(c)$ denote the fraction of mentions of concept c carrying*
 439 *a non-present assertion ($\alpha \neq \text{PRESENT}$). Without assertion labels, the maximum assertion-faithful*
 440 *accuracy for any downstream task conditioned on c is bounded by $1 - f_{np}(c)$. In clinical records*
 441 *where negated and uncertain mentions are prevalent—e.g., “no pneumonia,” “possible CHF”—this*
 442 *bound can be substantially below 1.*

443 *Proof.* Without assertion labels, any predictor must assign a single assertion class to all mentions
 444 of c . Choosing PRESENT (the majority class in clinical text) yields accuracy $1 - f_{np}(c)$; the $f_{np}(c)$
 445 non-present mentions are necessarily misclassified. $\square$

446 C Gap Analysis

447 Partial marks indicate limited coverage: GFM-RAG, MedRAG, and KARE build population-level
 448 or hierarchical KGs (not per-patient longitudinal graphs); KARE uses KG-augmented retrieval but

Table 7: Capability comparison across representative systems. ✓ = supported, ○ = partial, ✗ = not supported.

System	Patient KG	OMOP mapping	Assertion (7-class)	Tri-temporal	Assert-aware RAG	Experiencer	Allen’s algebra	Multi-hop (≥ 3)
DoctorRAG [15]	✗	✗	✗	✗	✗	✗	✗	✗
GFM-RAG [5]	○	✗	✗	✗	✗	✗	✗	✓
MedRAG [16]	○	✗	✗	✗	✗	✗	✗	○
KARE [6]	○	✗	✗	✗	○	✗	✗	✓
Multi-LLM KG [18]	✗	✓	○	✗	✗	✗	✗	✗
MedTKG [10]	✓	✗	✗	○	✗	✗	✗	✓
Graphiti [11]	✗	✗	✗	○	✗	✗	✗	✓
EpiKG (ours)	✓	✓	✓	✓	✓	✓	✓	○

without assertion awareness; Multi-LLM KG models uncertainty via entropy but not the full as-
 sersion taxonomy; MedTKG and Graphiti use temporal annotations but not tri-temporal models.
 EpiKG’s partial mark on multi-hop traversal (≥ 3 hops) reflects a deliberate architectural trade-
 off: PostgreSQL CTE-based traversal provides ACID compliance but degrades beyond 2 hops (Ap-
 pendix K).

D Safety Score

The safety score $S = 1 - \frac{1}{N} \sum_{i=1}^N w_i \cdot \mathbb{I}[\hat{y}_i \neq y_i]$ applies $w_i = 2.0$ for false-positive assertion
 errors (reporting a negated or absent condition as present) and $w_i = 1.0$ otherwise, capturing asym-
 metric clinical risk where acting on a falsely affirmed condition is more dangerous than missing a
 present one. We treat $w = 2.0$ as a reasonable default; sensitivity to this choice is a direction for
 future work.

E Cross-Model Per-Category Results

Table 8: ClinicalBench per-category accuracy (%) by model under C1 and C4g (intent-aware KG-RAG).
 KG-RAG benefit is model-dependent: Opus gains on all categories while MedGemma regresses on histori-
 cal (−46 pp).

Category	n	Claude Opus 4.6			MedGemma 27B		
		C1	C4g	Δ	C1	C4g	Δ
Negation	110	86.4	88.2	↑1.8 pp	93.6	86.4	↓7.3 pp
Conditional	20	15.0	45.0	↑30.0 pp	45.0	65.0	↑20.0 pp
Uncertainty	40	17.5	50.0	↑32.5 pp	10.0	32.5	↑22.5 pp
Family hist.	30	3.3	56.7	↑53.3 pp	3.3	56.7	↑53.3 pp
Sequence	40	100.0	100.0	=	79.4	90.0	↑10.6 pp
Current st.	50	40.0	72.0	↑32.0 pp	61.7	76.0	↑14.3 pp
Duration	30	86.7	93.3	↑6.7 pp	46.4	46.7	↑0.2 pp
Historical	50	6.0	62.0	↑56.0 pp	86.4	40.0	↓46.4 pp
Change	30	13.3	86.7	↑73.3 pp	11.5	60.0	↑48.5 pp
Overall	400	49.8	76.0	↑26.3 pp	59.9	66.0	↑6.1 pp

F MedGemma Full Per-Category Results

G Experiencer Attribute Propagation Ablation

The experiencer attribute distinguishes patient conditions from family member conditions;
 without it, the graph conflates the two, causing family history misattribution.

Category	n	C7	C6	C1	C4	C4g
Negation	110	13.6	72.7	93.6	85.5	86.4
Conditional	20	0.0	70.0	45.0	55.0	65.0
Uncertainty	40	0.0	32.5	10.0	30.0	32.5
Family History	30	0.0	63.3	3.3	40.0	56.7
Sequence	40	0.0	82.5	79.4	82.5	90.0
Current State	50	0.0	16.0	61.7	52.0	76.0
Duration	30	0.0	20.0	46.4	63.3	46.7
Historical	50	0.0	20.0	86.4	20.0	40.0
Change	30	0.0	10.0	11.5	10.0	60.0
Overall	400	3.8	46.5	59.9	55.0	66.0

Table 9: MedGemma 27B ClinicalBench per-category accuracy (% , deterministic keyword evaluator) for all five conditions. Historical regression (−46 pp, C1→C4g) contrasts with Opus’s +56 pp gain on the same category.

Table 10: Impact of experienter attribute propagation on Opus C4 (400 questions, prior evaluator run). Categories with no change omitted.

Category	Pre-fix	Post-fix	Δ
Family history	63.3%	73.3%	+10.0 pp
Uncertainty	50.0%	55.0%	+5.0 pp
Historical	34.0%	38.0%	+4.0 pp
Current state	46.0%	48.0%	+2.0 pp
Overall C4	68.2%	70.2%	+2.0 pp

465 The fix improved family history by +10.0 pp with zero regressions on guard categories (negation,
466 conditional, duration, sequence all unchanged), confirming that the experienter attribute is load-
467 bearing for assertion-sensitive reasoning.

468 H SliceBench Figures

469 I Per-Category Delta Chart

470 J SliceBench Conditions

Table 11: SliceBench conditions. B0–B4 form a monotone context progression.

ID	Condition	Context Provided to LLM
B0	LLM Alone	Question only (parametric knowledge)
B1	Latest Note	Most recent clinical note
B2	All Notes RAG	All notes, retrieved by relevance
B3	KG-RAG	Knowledge graph context + all notes
B4	Full System	KG-RAG + guidelines + calculators

471 K System Implementation Details

472 **LLM baseline.** On MedQA-USMLE (965 questions), Claude Opus 4.5 achieves 81.6% accu-
473 racy (5.1 pp below GPT-4 at 86.7% and Med-PaLM 2 at 86.5% [30]), establishing that the Claude
474 model family—Sonnet 4.5 (SliceBench answerer) and Opus 4.6 (ClinicalBench primary, SliceBench
475 judge)—provides a reasonable LLM baseline (Table 12).

476 **KG scale and traversal latency.** The deployed system processes 145 documents across 85 pa-
477 tients, materializing 3,100 KG nodes and 8,803 edges. Graph traversal latency is sub-millisecond
478 at this scale: 0.57 ms (1-hop), 0.75 ms (2-hop). The full system integrates 201 clinical calculators,
479 1,202 guideline sections, 20M+ OMOP vocabulary relationships, 101 assertion trigger patterns, and
480 24 edge types across 13 node types.

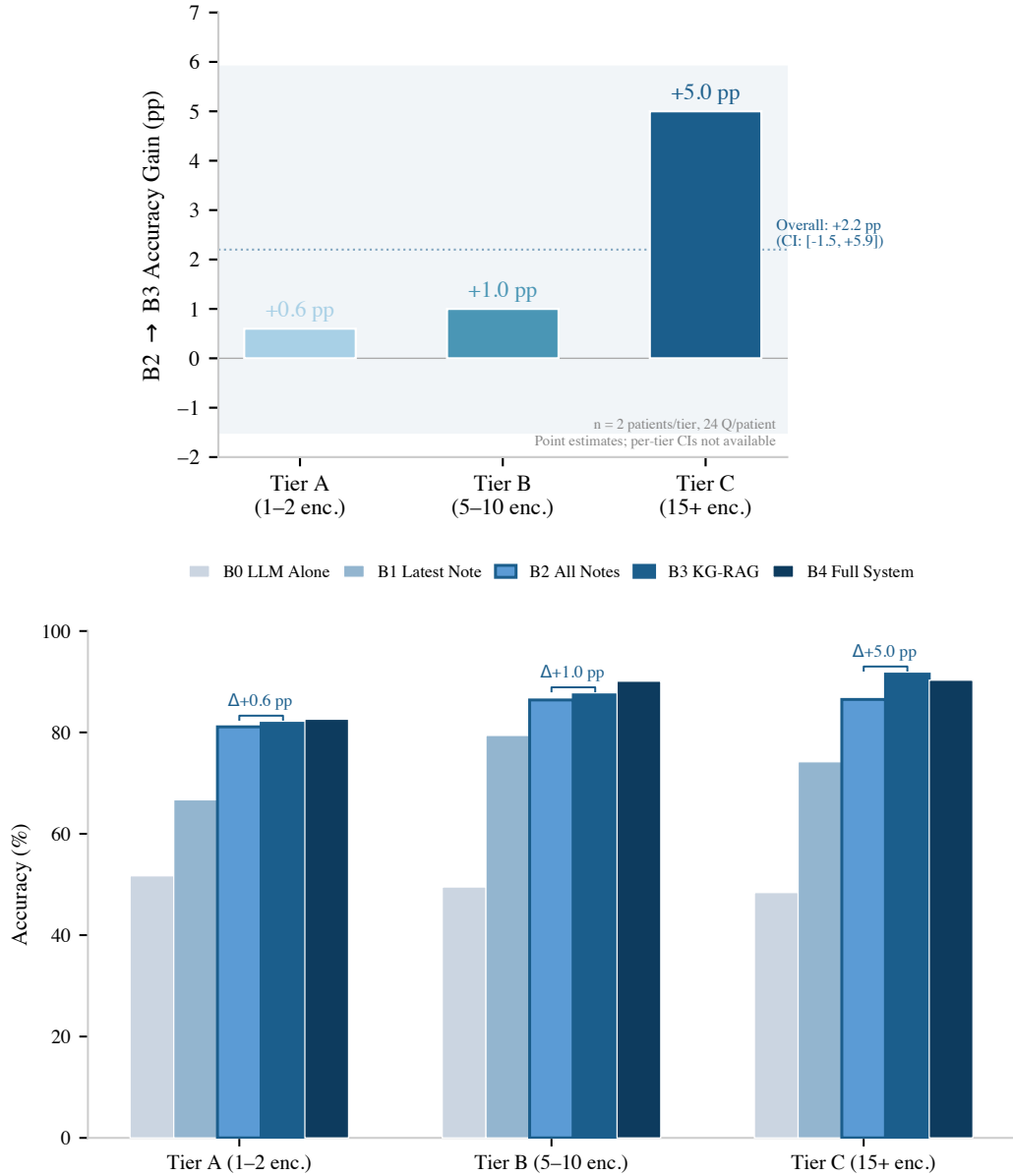


Figure 3: SliceBench results ($n = 2$ patients per tier, 24 questions each; point estimates without per-tier CIs). **Top:** The KG layer’s incremental benefit (B2→B3) shows a directional, complexity-dependent trend: +0.6 pp (Tier A) to +5.0 pp (Tier C). **Bottom:** Full condition progression by tier. The overall B2→B3 delta is +2.2 pp (CI: [-1.5 pp, +5.9 pp]).

Model	Accuracy	Step 1	N
EpiKG (LLM alone)	81.6%	79.9%	965
GPT-4 (2023)	86.7%	—	1,273
Med-PaLM 2 (2023)	86.5%	—	1,273
Claude 3 Opus (2024)	78.0%	—	1,273

Table 12: MedQA-USMLE results. Our LLM-alone baseline is competitive.

481 **Multi-hop traversal.** On the DR.KNOWS diagnostic reasoning benchmark [35], which measures
482 KG traversal accuracy using PostgreSQL-backed clinical data, EpiKG achieves 0.420 overall (50%
483 at 1-hop, 25% at 2-hop, 0% at 3-hop). Multi-hop degradation reflects a deliberate architectural

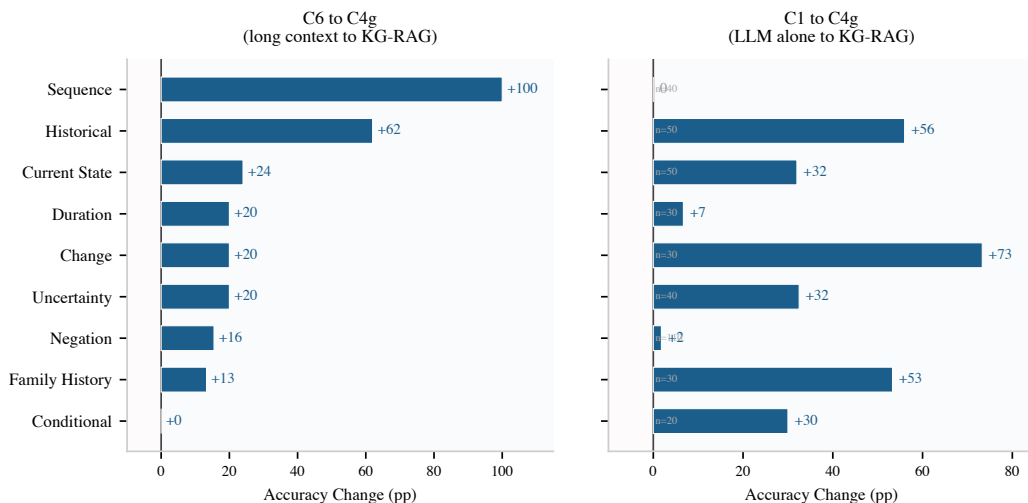


Figure 4: Per-category accuracy change (Claude Opus 4.6) for two comparisons. **Left:** C6→C4g (structured vs. unstructured retrieval): KG-RAG outperforms long context on all 9 categories, with the largest gains on sequence (+100 pp) and historical (+62 pp). **Right:** C1→C4g (LLM alone to KG-RAG): gains on 8 of 9 categories—change (+73 pp), historical (+56 pp), family history (+53 pp)—with zero regressions.

trade-off: PostgreSQL CTE-based traversal provides ACID compliance but scales poorly beyond 2 hops.

L SliceBench Tier-Stratified Results

Table 13: SliceBench results stratified by patient complexity tier. B2→B3 Δ shows the incremental KG contribution.

Tier	Encounters	Score (%)					B2→B3 Δ
		B0	B1	B2	B3	B4	
A	1–2 notes	51.7	66.7	81.1	81.8	82.6	+0.6 pp
B	5–10 notes	49.5	79.4	86.4	87.4	90.1	+1.0 pp
C	15+ notes	48.4	74.2	86.5	91.5	90.3	+5.0 pp

M ClinicalBench Full Per-Category Results (Opus)

Category	<i>n</i>	C7	C6	C1	C4	C4g
Negation	110	13.6	72.7	86.4	100.0	88.2
Conditional	20	0.0	45.0	15.0	70.0	45.0
Uncertainty	40	0.0	30.0	17.5	57.5	50.0
Family History	30	0.0	43.3	3.3	70.0	56.7
Sequence	40	0.0	0.0	100.0	87.5	100.0
Current State	50	0.0	48.0	40.0	54.0	72.0
Duration	30	0.0	73.3	86.7	83.3	93.3
Historical	50	0.0	0.0	6.0	60.0	62.0
Change	30	0.0	66.7	13.3	3.3	86.7
Overall	400	3.8	45.0	49.8	71.5	76.0

Table 14: Complete ClinicalBench per-category accuracy (%), Claude Opus 4.6, deterministic keyword evaluator) for all conditions. C7 is model-independent. C4→C4g: change jumps from 3.3% to 86.7%, isolating intent-aware routing.

N Human Validation Pilot

To assess LLM-as-judge reliability, we conducted a pilot validation study with a single board-certified physician who scored system outputs on a 4-point scale without knowledge of condition labels. The pilot ($n = 5$) revealed overly strict automated scoring on uncertainty and sequence questions. Gold standards were fully correct in 3/5 cases; model answers were partially correct or better in 3/5 cases where the auto-scorer marked them incorrect. A full two-physician blinded adjudication is preregistered (Section 5.5, Appendix O).

O Physician Validation Protocol

Design. Two board-certified emergency medicine physicians independently score ClinicalBench answers under two conditions (C1 and C4g) using the same binary correctness rubric as the deterministic keyword evaluator. Reviewers are blinded to condition assignment. The sample comprises 168 questions stratified across all 9 categories, with both conditions scored per question (336 total scored pairs).

Inter-rater agreement. Cohen’s κ is computed between the two reviewers on the full 336 scored pairs. A third board-certified physician adjudicates all disagreements.

Endpoints. (1) Human-keyword evaluator concordance rate (fraction of automated scores confirmed by physicians). (2) Human-adjudicated C4g vs. C1 accuracy delta. (3) Category-level human accuracy.

Disclosure. Both primary reviewers are co-authors. We report κ to demonstrate independent judgment and disclose the relationship in the main text.

Status. This protocol was designed before examining primary results. *Results are pending.*

P Reproducibility Details

Models. ClinicalBench primary ablation: Claude Opus 4.6 (claude-opus-4-20250514, keyword evaluator). Cross-model: MedGemma 27B (gemma3:27b, 4-bit GGUF) via Ollama; Qwen 35B; same evaluator. SliceBench: Claude Sonnet 4.5 (claude-sonnet-4-5-20250929) answers, Opus 4.6 judges. Temperature 0 throughout; 4-bit quantization introduces GPU non-determinism (± 10 pp run-to-run for MedGemma), controlled via within-run paired comparisons.

Evaluation. The deterministic keyword evaluator performs exact word-boundary matching (regex `\bKEYWORD\b`) against gold-standard assertion and temporal keywords. Per-category keyword sets: negation (“no,” “denies,” “absent,” “negative”), uncertainty (“possible,” “suspected,” “may,” “likely,” “consider”), temporal (“before,” “after,” “during,” “changed,” “new”), plus category-specific terms. Evidence preambles (echoed graph context) are stripped before matching.

Retrieval. C2’s vanilla RAG uses TF-IDF over chunked clinical notes (512-token chunks, 64-token overlap), retrieving the top-5 chunks by cosine similarity. ClinicalBench conditions map to SliceBench: C1 $\approx$ B0, C2 $\approx$ B2, C4 $\approx$ B3, C5 $\approx$ B4.

Bootstrap. $n = 2,000$ resamples, seed 42, BCa method [33].

Data. De-identified MIMIC-IV clinical notes accessed under a PhysioNet Credentialed Health Data Use Agreement (v1.5.0).

Compute. ClinicalBench Opus C1/C4g: ~ 22 min each (API); Opus C6: ~ 3.5 h (API); C7: < 1 min (no LLM); MedGemma C1–C4g: ~ 3.6 h (single GPU); MedGemma C6: ~ 1.5 h; SliceBench: ~ 2 h (API).

Evaluator evolution. Key refinements: (i) substring to word-boundary matching, (ii) evidence preamble stripping. All main-text numbers use the final version.

Q Assertion Category Definitions

R Temporal Relation Mapping

The nine temporal relations $\mathcal{R}$ used on KG edges are derived from Allen’s 13 canonical interval relations [12] by merging symmetric pairs.

Assertion	Definition	Example
PRESENT	Condition currently affirmed	“has diabetes”
ABSENT	Condition explicitly negated	“denies chest pain”
POSSIBLE	Condition suspected, not confirmed	“possible pneumonia”
CONDITIONAL	Contingent on specific circumstances	“if febrile, start antibiotics”
HYPOTHETICAL	Discussed as a scenario	“would need dialysis if. . .”
FAMILY_HISTORY	Attributed to a family member	“mother had breast cancer”
HISTORICAL	Previously true, not necessarily current	“former smoker”

Table 15: The 7-value assertion taxonomy, extending i2b2 by separating HISTORICAL from FAMILY_HISTORY.

$\mathcal{R}$ value	Allen source(s)	Meaning
BEFORE	Before, Meets	A ends before B starts
AFTER	After, Met-by	A starts after B ends
DURING	During	A entirely within B
CONTAINS	Contains	A entirely contains B
OVERLAPS	Overlaps, Overlapped-by	A and B partially overlap
STARTS	Starts, Started-by	A and B share start time
FINISHES	Finishes, Finished-by	A and B share end time
CONCURRENT	Equals	A and B have same interval
UNKNOWN	—	Relation undetermined

Table 16: Mapping from Allen’s 13 interval relations to the 9 temporal relation values stored on KG edges.

S Intent-Aware Routing Algorithm

The C4g classifier is rule-based, mapping questions to one of four intents using keyword patterns (e.g., “changed”, “new since” trigger CHANGE; “currently”, “active problem” trigger CURRENT_STATE). Algorithm 1 details the full retrieval procedure.

Algorithm 1: Intent-Aware Retrieval (C4g)

Input: Question q , patient π
Output: Structured evidence E

```

1  $\mathcal{C} \leftarrow \text{EXTRACTCONCEPTS}(q)$ ; // NLP + OMOP enrichment
2  $\iota \leftarrow \text{CLASSIFYINTENT}(q)$ ; //  $\in \{\text{CHANGE, CURRST, HIST, DEFAULT}\}$ 
3 if  $\iota = \text{CHANGE}$  then
4   Partition edges by admission:  $\mathcal{E}_k \leftarrow \{e \mid \text{hadm\_id}(e) = k\}$ ;
5   foreach admission pair  $(k, k')$  with  $k < k'$  do
6      $\mathcal{A} \leftarrow \mathcal{C}_{k'} \setminus \mathcal{C}_k$ ;  $\mathcal{R} \leftarrow \mathcal{C}_k \setminus \mathcal{C}_{k'}$ ;  $\mathcal{S} \leftarrow \mathcal{C}_k \cap \mathcal{C}_{k'}$ ;
7      $E_g \leftarrow \text{FORMATCHANGE}(\mathcal{A}, \mathcal{R}, \mathcal{S})$ ;
8 else if  $\iota = \text{CURRST}$  then
9    $E_g \leftarrow \text{FILTEREDGES}(\pi, \mathcal{C}, \tau_a = \text{CURRENT} \vee \text{open validity})$ ;
10  Deduplicate by concept; emit “NOT FOUND” for missing  $c \in \mathcal{C}$ ;
11 else if  $\iota = \text{HIST}$  then
12   $E_g \leftarrow \text{FILTEREDGES}(\pi, \mathcal{C}, \tau_a = \text{PAST})$ ;
13  Augment: concepts in earlier admissions but absent from latest  $\rightarrow$  “resolved”;
14 else
15   $E_g \leftarrow \text{BIDIRECTIONALBFS}(\pi, \mathcal{C}, \text{hops} = 2-3, c_{\min} = 0.3)$ ;
16  $E_d \leftarrow \text{RETRIEVEDOCUMENTS}(\pi, \mathcal{C})$ ;
17 return  $\text{COMPOSE}(E_g, E_d, \text{template}(\iota))$ ;

```

T Bookend and Diagnostic Conditions

C5 (Full System). C5 extends C4 with guideline retrieval (1,202 sections) and clinical calculators (201, e.g., CHA₂DS₂-VASc, MELD). In a prior evaluator run, C5 achieved 68.2% overall, below

the prior-run C1 and C4, likely because non-relevant components dilute the context window.

C6 (Long Context). All patient documents are concatenated chronologically and presented to the LLM. For Opus (200K token window), all notes fit; for MedGemma (8K window), later documents are truncated. C6 achieves 45.0% (Opus) and 46.5% (MedGemma) overall. Both models catastrophically fail on historical (0% Opus, 20% MedGemma), confirming that brute-force long context cannot substitute for structured retrieval regardless of context window size.

C7 (Deterministic KG). KG edges matching query concepts are returned directly without LLM reasoning. C7 achieves 3.8% overall (only negation at 13.6% via stored assertion metadata), confirming that structured data without an LLM reasoner cannot answer clinical questions.

U Illustrative Retrieval Example

To illustrate why intent-aware routing matters, consider a change question: “*What medications changed between this patient’s first and second admissions?*”

C1 (LLM alone). The model sees only the current note, which mentions metoprolol, atorvastatin, and “discontinued lisinopril.” Without prior admission records, it cannot determine what was *added* vs. *continued*, and may hallucinate prior medication lists.

C4g (intent-aware KG-RAG). The intent classifier routes to CHANGE retrieval, which partitions KG edges by `hadm_id` and computes set differences:

- **Added** (admission 2 only): atorvastatin 40mg [PRESENT]
- **Removed** (admission 1 only): lisinopril 10mg [PRESENT→ABSENT]
- **Continued**: metoprolol 25mg [PRESENT, both admissions]

The assertion labels (PRESENT, ABSENT) disambiguate: “discontinued lisinopril” is not a current medication but a historical one whose status changed—exactly the distinction the epistemic schema preserves.

V Ethics Statement

This work uses de-identified clinical data from MIMIC-IV [31] under a PhysioNet Credentialed Health Data Use Agreement. No patient re-identification was attempted. The system is designed for clinical decision *support*, not autonomous clinical decision-making.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: **Yes.**

Justification: The abstract and introduction state four specific contributions (Section 1), each supported by empirical evidence in Sections 5.1–5.4. SliceBench claims include bootstrap CIs; ClinicalBench claims use within-run paired comparisons (Section 4.3). Claims are scoped with “to our knowledge” qualifiers where appropriate.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: **Yes.**

Justification: Section 6 explicitly discusses limitations including small sample sizes (6 patients for SliceBench, 2 per tier), single-site data (MIMIC-IV), the B2→B3 confidence interval crossing zero, model-dependent effects, and pending physician adjudication.

3. Theory assumptions and proofs

585 Question: For each theoretical result, does the paper provide the full set of assumptions and a
586 complete (correct) proof?

587 Answer: **Yes.**

588 Justification: Proposition 3 (assertion entropy loss) and Corollary 4 (faithfulness bound) in
589 Appendix B state all assumptions explicitly. These are information-theoretic characterizations
590 with all steps included.

591 4. Experimental result reproducibility

592 Question: Does the paper fully disclose all the information needed to reproduce the main
593 experimental results of the paper to the extent that it affects the main claims and/or conclusions
594 of the paper?

595 Answer: **Yes.**

596 Justification: Appendix P reports exact model versions (MedGemma 27B, Claude Sonnet 4.5,
597 Claude Opus 4.6), temperature settings (0), bootstrap parameters ($n = 2,000$, seed 42, BCa
598 method), and data source (MIMIC-IV v1.5.0 under PhysioNet DUA). Section 4 describes
599 question generation and evaluation protocols.

600 5. Open access to data and code

601 Question: Does the paper provide open access to the data and code, with sufficient instructions
602 to faithfully reproduce the main experimental results?

603 Answer: **No.**

604 Justification: MIMIC-IV data is available under PhysioNet credentialed access (not open).
605 Code will be released upon acceptance. Benchmark question templates and evaluation scripts
606 will be included in the release.

607 6. Experimental setting/details

608 Question: Does the paper specify all the training and test details necessary to understand the
609 results?

610 Answer: **Yes.**

611 Justification: Section 4 specifies benchmark design, question counts, condition definitions,
612 patient selection criteria, and evaluation methodology. Appendix P provides model versions
613 and computational requirements.

614 7. Experiment statistical significance

615 Question: Does the paper report error bars suitably and correctly?

616 Answer: **Yes.**

617 Justification: Table 5 reports BCa bootstrap 95% CIs for all conditions. Per-tier results (Fig-
618 ure 3) are clearly labeled as point estimates without per-tier CIs due to small tier-level sample
619 sizes ($n = 2$ patients per tier). The overall B2→B3 CI crossing zero is explicitly noted.

620 8. Experiments compute resources

621 Question: For each experiment, does the paper provide sufficient information on the computer
622 resources needed to reproduce the experiments?

623 Answer: **Yes.**

624 Justification: Appendix P reports computational requirements: ClinicalBench Opus $\approx$ 22 min
625 per condition (API), MedGemma $\approx$ 3.6 hours (single GPU), SliceBench $\approx$ 2 hours (API).

626 9. Code of ethics

627 Question: Does the research conducted in the paper conform, in every respect, with the
628 NeurIPS Code of Ethics?

629 Answer: **Yes.**

630 Justification: The study uses de-identified clinical data under an approved data use agreement
631 (Appendix V). No patient re-identification was attempted. The system is designed for clinical
632 decision *support*, not autonomous decision-making.

633 10. Broader impacts

634 Question: Does the paper discuss both potential positive societal impacts and negative societal
635 impacts of the work?

636 Answer: **Yes.**

637 Justification: Section 6 discusses the potential for improved clinical decision support (positive)
638 and risks of over-reliance on automated systems, context dilution, and the need for physician
639 oversight (negative). Appendix V addresses data ethics.

640 11. Safeguards

641 Question: Does the paper describe safeguards that have been put in place for responsible
642 release of data or models?

643 Answer: **Yes.**

644 Justification: The system operates on de-identified data under PhysioNet DUA. It is designed
645 as a decision support tool requiring physician oversight. Code release will include usage
646 guidelines emphasizing the system is not validated for autonomous clinical use.

647 12. Licenses for existing assets

648 Question: Are the creators or original owners of assets used in the paper properly credited and
649 are the license and terms of use explicitly mentioned and properly respected?

650 Answer: **Yes.**

651 Justification: MIMIC-IV is cited [31] and used under PhysioNet Credentialed Health Data
652 Use Agreement. All benchmark frameworks and models are cited. OMOP CDM is an open
653 standard.

654 13. New assets

655 Question: Are new assets introduced in the paper well documented and is the documentation
656 provided alongside the assets?

657 Answer: **Yes.**

658 Justification: ClinicalBench and SliceBench are new benchmarks fully documented in Sec-
659 tion 4, with question generation procedures, category definitions (Appendix Q), and evalua-
660 tion protocols. Templates and scoring scripts will accompany the code release.

661 14. Crowdsourcing and research with human subjects

662 Question: For crowdsourcing experiments and research with human subjects, does the paper
663 include the full text of instructions given to participants and screenshots?

664 Answer: **N/A.**

665 Justification: No crowdsourcing was used. Physician adjudication (Appendix [N](#)) involves
666 expert review by co-authors, not human subjects research.

667 **15. IRB approvals or equivalent**

668 Question: Does the paper describe potential risks and does it include the IRB approval number
669 if applicable?

670 Answer: **N/A.**

671 Justification: MIMIC-IV is a de-identified public dataset; use under PhysioNet DUA does not
672 require separate IRB approval per PhysioNet policy.